14	**(a)**		$\frac{16}{x^2}+\frac{3}{x}=0$	**M1**	**1.1**	sets equal to 0 and obtains value(s) for x
			$(-\frac{16}{3}, 0)$ and no others	**A1** **[2]**	**1.1**	allow (–5.3, 0) or better; allow eg $x = -5.3$, $y = 0$ if **M0** allow **SC1** for **exact** coordinates unsupported
14	**(b)**	**(i)**	$\left[\frac{dy}{dx}\right] = -2 \times 16x^{-3} - 3x^{-2}$ **oe**	**M1***	**1.1**	allow one sign error in coefficient;
			$\left[\frac{dy}{dx}\right] = -32x^{-3} - 3x^{-2}$ **or** $-\frac{32}{x^3}-\frac{3}{x^2}$ **isw**	**A1** **[2]**	**1.1**	both terms correct and no extras
	(b)	**(ii)**	their $-32x^{-3} - 3x^{-2} = 0$ used to obtain value(s) for x	**M1dep***	**1.1**	
			$\left(-\frac{32}{3}, -\frac{9}{64}\right)$ **or** $\left(-10\frac{2}{3}, -\frac{9}{64}\right)$ **cao**	**A1** **[2]**	**1.1**	allow eg -0.140625; allow eg $x = -\frac{32}{3}, y = -\frac{9}{64}$
14	**(c)**	**(i)**	$\left[\frac{d^2y}{dx^2}\right] = 96x^{-4} + 6x^{-3}$ **or** $\frac{96}{x^4}+\frac{6}{x^3}$	**M1**	**1.1**	differentiation of their $\frac{dy}{dx}$, dependent on award of **M1** in **14 (b)(i)**; allow one sign error in coefficient
				A1 **[2]**	**1.1**	both terms correct

Question			Answer	Marks	AO	Guidance
14	(c)	(ii)	$96x^{-4} + 6x^{-3} < 0$	M1	2.1	putting their expression for $\frac{d^2y}{dx^2} < 0$ or $\frac{d^2y}{dx^2} = 0$ or $\frac{d^2y}{dx^2} \leq 0$ to find value(s) for x
			$x < -16$ or $-16 > x$ **cao**	A1	1.1	
				[2]		
14	(d)		$\int \frac{16}{x^2} + \frac{3}{x} dx = -\frac{16}{x} + 3\ln\|x\|$ **oe**	M1*	1.1	one term correct; allow omission of modulus sign
				A1	2.1	both terms correct; ignore + c; allow omission of modulus sign
			$\left[\frac{-16}{-½} + 3\ln\|-½\|\right] - \left[\frac{-16}{-4} + 3\ln\|-4|\right]$	M1dep*	1.1	allow omission of modulus sign; allow one incorrect sign but do not allow eg $F\left[-\frac{1}{2}\right] + F[-4]$
			$3\ln\|-½\| = 3\ln\frac{1}{2}$ and $3\ln\|-4\| = 3\ln 4$	M1dep*	2.1	**oe;** modulus seen and used to deal with both terms
			$28 - 9\ln 2$ **cao**	A1	1.1	may see eg $32 - 4 - 3\ln 2 - 6\ln 2$ **oe** as intermediate step
				[5]		
14	(d)		*Alternatively* $\int_{-4}^{-\frac{1}{2}} \frac{16}{x^2} + \frac{3}{x} dx = \int_{\frac{1}{2}}^{4} \frac{16}{x^2} - \frac{3}{x} dx$	M1*		by reflecting in y-axis and working with $f(-x)$ **oe;** may be awarded before first **M1;** condone incorrect reason eg reflection in x-axis; **M0** for use of $\int_{\frac{1}{2}}^{4} \frac{16}{x^2} + \frac{3}{x} dx$ by symmetry
			$\int \frac{16}{x^2} + \frac{3}{x} dx = -\frac{16}{x} - 3\ln x$ **oe**	M1*		one term correct; allow omission of modulus sign
				A1		both terms correct; ignore + c; allow omission of modulus sign
			$\left[\frac{-16}{4} - 3\ln 4\right] - \left[\frac{-16}{\frac{1}{2}} - 3\ln\frac{1}{2}\right]$	M1dep*		dependent on correct integration and correct transformation
			$28 - 9\ln 2$ **cao**	A1		may see eg $32 - 4 - 3\ln 2 - 6\ln 2$ **oe** as intermediate step
				[5]		